

Cauchy Duals for Uniform Measures on Regular Polygons

An exact rank dichotomy extending the three-point example

Candidate partial result for arXiv:2511.04565

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Status: candidate partial proof, likely valid. The result completely classifies the uniform regular-polygon family, but does not settle arbitrary three-point support and arbitrary positive masses. Novelty remains subject to expert literature review.

1 Source question and scope

Joshi, Phatak, and Sholapurkar study the Cauchy dual of multiplication by z on Dirichlet-type spaces associated with finitely supported measures [1]. Their epilogue conjecture appears on source PDF p. 10:

It can be seen that $(\alpha^2 \overline{w})$ is not a root of the quadratic equation $\left((1-b) + \frac{3b}{x+1}\right)Y^2 + \frac{3b}{x(x+1)}Y + \frac{3b}{x(x-1)} = 0$ implying that $\sum_{j=1}^3 p_j(\alpha) \overline{p_j(\alpha w)} \neq 0$. Therefore, the Cauchy dual M'_z of M_z on $D(\mu)$ is not subnormal. $\square$

4. EPILOGUE

The authors believe that the main theorem of the present article can be proved for the measure μ in the general form, i.e. $\mu = c_1 \delta_{\zeta_1} + c_2 \delta_{\zeta_2} + c_3 \delta_{\zeta_3}$, where c_1, c_2, c_3 are positive real numbers, but the computational complexity increases substantially. Further, there is a strong evidence that the following conjecture has an affirmative solution.

Conjecture. *If $\zeta_1, \zeta_2, \zeta_3$ are any three distinct points on the unit circle and if $\mu = c_1 \delta_{\zeta_1} + c_2 \delta_{\zeta_2} + c_3 \delta_{\zeta_3}$, then the Cauchy dual M'_z of M_z on $D(\mu)$ is not subnormal.*

DECLARATIONS

The present work is carried out at the research center at the Department of Math-

Exact transcription. “If $\zeta_1, \zeta_2, \zeta_3$ are any three distinct points on the unit circle and if $\mu = c_1 \delta_{\zeta_1} + c_2 \delta_{\zeta_2} + c_3 \delta_{\zeta_3}$, then the Cauchy dual M'_z of M_z on $D(\mu)$ is not subnormal.”

The paper proves this only for three equispaced points with all masses equal to one. We prove the common-mass case for every positive mass and, more generally, obtain a sharp dichotomy for uniform measures on regular N -gons. Thus this is a solved symmetric subfamily of the conjecture and a substantial contribution to the broader characterization problem stated on source PDF p. 4:

$$B_n(T) \leq 0 \text{ for all integers } n \geq 1.$$

Now we are ready to state and describe the problem that has been tackled in this article.

2.1. The Problem. We begin our discussion with the fact [7, Theorem 6.1] which states that for a cyclic, analytic 2-isometry T in $\mathcal{B}(\mathcal{H})$, the rank of $T^*T - I$ is finite if and only if there exist a finitely supported measure μ on the unit circle $\mathbb{T}$ such that T is unitarily equivalent to a multiplication by the coordinate function z on $D(\mu)$. In fact, if the rank of $T^*T - I$ is a positive integer k , then the corresponding measure μ is supported at exactly k points of unit circle $\mathbb{T}$. In view of this result, a construction of a cyclic, analytic 2-isometry reduces to choosing finitely many points on the unit circle and looking at the multiplication operator M_z on the Dirichlet space $D(\mu)$, where μ is supported at the chosen points.

We now state the problem under consideration:

Problem 1. *Characterize finitely supported, positive, Borel measures μ on unit circle $\mathbb{T}$ such that the Cauchy dual of M_z on $D(\mu)$ is subnormal.*

2 Definitions, criterion, and proof intuition

For a finite positive Borel measure μ on $\mathbb{T}$, let $\mathcal{D}(\mu)$ be the Dirichlet-type space with norm

$$\|f\|_{\mathcal{D}(\mu)}^2 = \|f\|_{H^2}^2 + \int_{\mathbb{D}} |f'(z)|^2 P_\mu(z) \, dA(z).$$

Multiplication $\mathcal{M}_z f = zf$ is a cyclic analytic 2-isometry. Its Cauchy dual is $\mathcal{M}'_z = \mathcal{M}_z(\mathcal{M}_z^* \mathcal{M}_z)^{-1}$.

We use the rational de Branges–Rovnyak model of Chavan–Ghara–Reza and Costara [2, 3]. In the notation needed below, $\mathcal{D}(\mu) = \mathcal{H}(B)$ with equality of norms, where

$$B = (p_1/q, \dots, p_N/q), \quad G(z, w) := \sum_{j=1}^N p_j(z) \overline{p_j(w)},$$

and the roots α_r of q lie outside $\overline{\mathbb{D}}$.

Lemma 1 (Forbidden-atom necessary condition). *Suppose q has distinct roots $\alpha_0, \dots, \alpha_{N-1}$. If the Cauchy dual of $\mathcal{M}_z$ on $\mathcal{H}(B)$ is subnormal, then the complex atomic measure*

$$\nu = \sum_{r,t=0}^{N-1} \frac{G(\alpha_r, \alpha_t)}{\alpha_r^2 \overline{\alpha_t}^2 q'(\alpha_r) \overline{q'(\alpha_t)}} \delta_{1/(\alpha_r \overline{\alpha_t})}$$

is a positive measure supported on $[0, 1]$.

This is the necessary-measure consequence of the rational-symbol criterion in [2, Corollary 4.2 and Remark 4.3]. It is important here that equal pole products are grouped as atoms of one measure. Briefly, subnormality makes the corresponding scalar sequence completely monotone, so it is a Hausdorff moment sequence of a positive measure on $[0, 1]$. Partial fractions give the displayed finite complex measure with the same moments; polynomial approximation on the compact union of the two supports makes the representing measure unique. Hence every grouped coefficient at a point outside $[0, 1]$ must vanish.

Proof Intuition

For a regular polygon, root-of-unity orthogonality turns Costara’s spectral factor into the binomial $q(z) = z^N - \rho$. The auxiliary Gram matrix is circulant, so the discrete Fourier basis diagonalizes it. This yields a closed formula

$$G(z, w) = P(z\overline{w}) = \sum_{k=1}^N A_k (z\overline{w})^k$$

with every $A_k > 0$. Subnormality would force P to vanish at all $N - 1$ rotated off-diagonal pole products. Those prescribed roots determine $P(x)/x$ up to scale, but its first two coefficients disagree with the explicit A_k whenever $N \geq 3$. The exceptional equality at $N = 2$ is exactly the known antipodal subnormal case.

3 The regular-polygon dichotomy

Theorem 2. *Let $N \geq 1$, $c > 0$, $\xi \in \mathbb{T}$, and $\omega = e^{2\pi i/N}$. Set*

$$\mu_{N,c,\xi} = c \sum_{j=0}^{N-1} \delta_{\xi\omega^j}.$$

The Cauchy dual of multiplication by z on $\mathcal{D}(\mu_{N,c,\xi})$ is subnormal if and only if $N \leq 2$.

Proof. Rotation of the measure gives unitary equivalence up to multiplication of the operator by a unimodular scalar [2, Proposition 7.1], so take $\xi = 1$. Write $\zeta_j = \omega^j$ and $s(z) = z^N - 1$. On $\mathbb{T}$, root-of-unity orthogonality gives

$$\sum_{j=0}^{N-1} \left| \frac{s(z)}{z - \zeta_j} \right|^2 = N^2.$$

Consequently Costara's trigonometric polynomial is

$$|s(z)|^2 + c \sum_j \left| \frac{s(z)}{z - \zeta_j} \right|^2 = 2 + cN^2 - z^N - \bar{z}^N = \rho^{-1} |z^N - \rho|^2, \quad (1)$$

where $\rho > 1$ is determined by

$$\rho + \rho^{-1} = 2 + cN^2, \quad c = \frac{(\rho - 1)^2}{N^2\rho}. \quad (2)$$

Thus we may take

$$q(z) = z^N - \rho, \quad O(z) = \sqrt{\rho} \frac{z^N - 1}{z^N - \rho}.$$

We now compute the numerator kernel G . Costara's formula uses

$$f_j(z) = \frac{O(z)}{O'(\zeta_j)(z - \zeta_j)} = \frac{1 - \rho}{N} \frac{\sum_{m=0}^{N-1} \zeta_j^{-m} z^m}{q(z)}. \quad (3)$$

Let $\Gamma = (\langle f_i, f_j \rangle)$. The formulas in [3, Lemmas 4.2–4.3], together with (2), give

$$\Gamma_{ii} = c \left(\frac{N-1}{2} + \frac{N}{\rho-1} \right), \quad \Gamma_{ij} = c \frac{\zeta_i/\zeta_j}{1 - \zeta_i/\zeta_j} \quad (i \neq j). \quad (4)$$

For $u_m = (\zeta_j^{-m})_{j=0}^{N-1}$, the elementary root-of-unity sum

$$\sum_{k=1}^{N-1} \frac{\omega^{(m+1)k}}{1 - \omega^k} = m - \frac{N-1}{2}$$

(with a harmless reversal of indices) shows that

$$\Gamma u_m = c \left(\frac{N}{\rho-1} + m \right) u_m, \quad 0 \leq m < N. \quad (5)$$

Since the u_m are orthogonal and have squared norm N , (3)–(5) yield

$$q(z)\overline{q(w)} \sum_{i,j} (\Gamma^{-1})_{ji} f_j(z) \overline{f_i(w)} = \sum_{m=0}^{N-1} h_m (z\overline{w})^m, \quad (6)$$

where

$$h_m = \frac{N\rho(\rho-1)}{N+m(\rho-1)}. \quad (7)$$

The reproducing-kernel identity in [2, Theorem 6.4] says that

$$G(z, w) = q(z)\overline{q(w)} - h\overline{o(s(z)\overline{s(w)})} - (1 - z\overline{w}) \sum_{m=0}^{N-1} h_m (z\overline{w})^m.$$

The first two terms reduce to $(\rho-1)(\rho-(z\overline{w})^N)$. Since $h_{m-1} - h_m$ telescopes, we obtain the exact diagonal polynomial

$$G(z, w) = P(z\overline{w}), \quad P(x) = \sum_{k=1}^N A_k x^k, \quad (8)$$

with

$$A_k = \frac{N\rho(\rho-1)^2}{[N+(k-1)(\rho-1)][N+k(\rho-1)]} > 0. \quad (9)$$

Let $a = \rho^{1/N}$. The roots of q are $\alpha_r = a\omega^r$. For a nonzero residue $d \pmod{N}$, all N ordered pairs with $r - t = d$ have the same product $x_d = \alpha_r \overline{\alpha_t} = a^2 \omega^d$, and $q'(\alpha_r) = N\alpha_r^{N-1}$. Hence the coefficient of the grouped atom at $1/x_d$ in Lemma 1 is

$$C_d = \frac{P(x_d)}{N\rho^2 x_d}. \quad (10)$$

Because $d \neq 0$, $1/x_d \notin [0, 1]$. If the Cauchy dual were subnormal, Lemma 1 would therefore force

$$P(a^2 \omega^d) = 0, \quad d = 1, \dots, N-1. \quad (11)$$

Assume $N \geq 3$. The polynomial $R(x) = P(x)/x$ has degree $N-1$. If (11) held, its $N-1$ distinct roots would give

$$R(x) = A_N \prod_{d=1}^{N-1} (x - a^2 \omega^d) = A_N \frac{x^N - a^{2N}}{x - a^2}.$$

Comparing its constant and linear coefficients yields $A_2/A_1 = a^{-2} = \rho^{-2/N}$. But (9) gives

$$\frac{A_2}{A_1} = \frac{N}{N+2(\rho-1)}. \quad (12)$$

These cannot agree. Indeed, with $a > 1$, agreement is equivalent to

$$a^2 = 1 + \frac{2(a^N - 1)}{N}.$$

The right side minus the left side vanishes at $a = 1$ and has derivative $2a(a^{N-2} - 1) > 0$ for $a > 1$ and $N > 2$. This contradiction proves non-subnormality for every $N \geq 3$.

For $N = 1$, subnormality is the established one-atom theorem [2, Corollary 5.4]. For $N = 2$, the two support points are antipodal, and the established antipodal theorem applies for arbitrary positive masses [2, Theorem 2.4]. This proves the dichotomy. $\square$

Corollary 3. *For every $c > 0$, the Cauchy dual on $\mathcal{D}(c\delta_1 + c\delta_\omega + c\delta_{\omega^2})$, where $\omega^3 = 1$ and $\omega \neq 1$, is not subnormal.*

This is the source theorem with the common mass upgraded from $c = 1$ to every $c > 0$. Theorem 2 simultaneously extends it to every regular polygon of rank at least three.

4 Verification, limitations, and novelty

The accompanying verifier independently constructs the circulant Gram matrix, inverts it numerically, reconstructs the finite kernel, and checks (8)–(9) for $N = 2, 3, 4, 5, 8$ and several positive masses. It also checks that the forbidden values vanish for $N = 2$ and fail to vanish for every tested $N \geq 3$. These computations are sanity checks; the proof above is analytic.

Scope limitation. The arbitrary three-point/three-weight conjecture remains open here. Unequal weights destroy the circulant structure, while arbitrary support also destroys the binomial spectral factor. Eight focused upgrade attempts included direct general three-atom reconstruction, optimization searches for exceptional subnormal candidates, and symbolic work on unequal equilateral weights; none yielded a responsible full proof or counterexample.

Bounded novelty audit. The run indexes and primary-source arXiv searches through 13 August 2026 were checked using arXiv:2103.10059, 2510.14004, and 2511.04565 and the phrases “Cauchy dual subnormality,” “three distinct points,” “rank three,” “regular N -gon,” “equi-spaced points,” and “uniform measure.” The search found the source, the rational-symbol criterion, and rank-two results, but no statement of the uniform regular- N -gon dichotomy. Novelty is plausible, not certified.

Human-review recommendation: send to an operator theorist. The highest-value checks are the conjugation convention in (6), the DFT eigenvalue indexing in (5), and the grouping calculation (10). All three are also covered by the independent numerical reconstruction.

References

- [1] S. A. Joshi, G. M. Phatak, and V. M. Sholapurkar, *An example of a cyclic analytic 2-isometry with defect operator of rank 3, whose Cauchy dual is not subnormal*, arXiv:2511.04565 (2025).
- [2] S. Chavan, S. Ghara, and M. R. Reza, *The Cauchy dual subnormality problem via de Branges–Rovnyak spaces*, *Studia Math.* **265**(3) (2022); arXiv:2103.10059.
- [3] C. Costara, *Reproducing kernels for Dirichlet spaces associated to finitely supported measures*, *Complex Anal. Oper. Theory* **10** (2016), 1277–1293.